4.1: General Flexure Formula

- **In establishing these equations, we assumed that line $\overline{CS}$ (which was along the x axis) underwent no deformation**

- **So in drawing our coordinate frame, we need to position the x axis such that it intersects the neutral axis (if such a thing exists?)**

- **We're familiar with the concept of a neutral axis when bending is strictly about the z axis; what about in these more general scenarios?**

- **We know that zero deformation implies $\varepsilon_x = 0$ which in turn (via Hooke's Law) implies $\sigma_x = 0$**

- **From $\sigma_x = -EB_1 y + EB_2 z$ we see that $\sigma_x = 0$ occurs at every point for which $y = \frac{B_2}{B_1} z$**

4.1: General Flexure Formula

- Therefore a neutral axis does exist that forms a line across the cross-section that makes an angle of $tan^{-1}\left(\frac{B_2}{B_1}\right)$ with the horizontal

- This establishes the slope of the neutral axis, but we still need to establish the "intercept," i.e. can we locate a specific point on the cross-section through which the neutral axis passes?

- Since we assume that this stress arose due to pure bending, the total net force at each cross-section (i.e. the sum of all differential forces arising from stress acting on each differential area of the cross-section) must be 0 → $\int_A \sigma_x dA = 0$

- Substituting $\sigma_x = -EB_1y + EB_2z$ yields $\int_A (-EB_1y + EB_2z)dA = 0$

- $\rightarrow EB_2\int_A zdA - EB_1\int_A ydA = 0 \rightarrow B_2\int_A zdA - B_1\int_A ydA = 0$

4.1: General Flexure Formula

- **Note that $\int_A zdA$ and $\int_A ydA$ are expressions for the first moment of area about the y and z axes, respectively (we typically label this *"Q"*)**

- **We know that $\int_A zdA = A\bar{z}$ where A is the cross-section area and $\bar{z}$ is the distance from the y axis to the centroid (i.e. the z component of the centroid); similarly $\int_A ydA = A\bar{y}$**

- **So the last equation on the previous slide becomes $B_2\bar{z} - B_1\bar{y} = 0$**

- **This is the same as the equation for the neutral axis, from which we conclude that the neutral axis must pass through the centroid, i.e. the centroid must lie on the neutral axis (this makes logical sense!)**

4.1: General Flexure Formula

- **So in order for the General Flexure Formula (in the way we have derived it) to hold, the x axis must intersect the neutral axis, which in turn must pass through the centroid**

- **For a given cross section, this is also the point where our y and z axes will intersect (i.e. origin of our chosen coordinate frame)**

- **Since the line on which this point may lie will vary from case to case depending on the values of B_1 and B_2, the most logical choice to place the origin is at the centroid**

- **In this case we have $\bar{y} = \bar{z} = 0$ regardless the values of B_1 and B_2!**

4.1: General Flexure Formula

- **Now that we have the General Flexure Formula, we can easily see how the equation simplifies in certain circumstances:**
  - When $M_y = 0$, $\sigma_x = \frac{(-yI_y + zI_{yz})M_z}{I_y I_z - {I_{yz}}^2}$
  - When $M_z = 0$, $\sigma_x = \frac{(-yI_{yz} + zI_z)M_y}{I_y I_z - {I_{yz}}^2}$
  - When $I_{yz} = 0$, $\sigma_x = \frac{-yI_y M_z + zI_z M_y}{I_y I_z}$
  - When $M_z = 0$ and $I_{yz} = 0$, $\sigma_x = \frac{zM_y}{I_y}$
  - When $M_y = 0$ and $I_{yz} = 0$, $\sigma_x = -\frac{yM_z}{I_z}$
- **The last case above is the one we are most familiar with (see Slide 3!)**

4.1: General Flexure Formula

- **Note that on slide 3 we stated that the assumptions in order $\sigma_x = -\frac{yM_z}{I_z}$ to hold are M_y = 0 and a symmetric cross-section**

- **In fact, the 2nd assumption only needs to be I_{yz} = 0, i.e. that the y and z axes must be aligned with the principal axes of the cross-section**

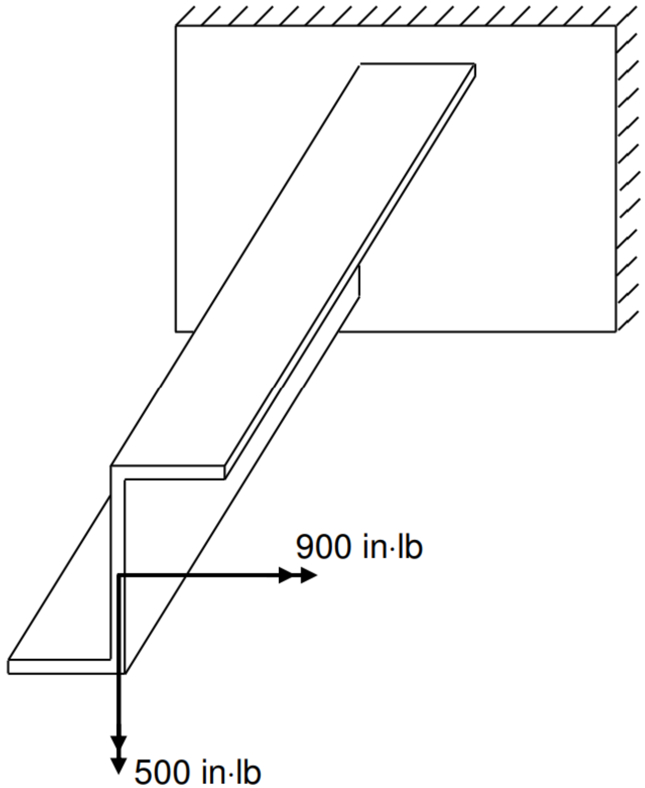
- **Another way of stating this is that the applied moment (M_z) must be directed along one of the principal axes of the cross-section**

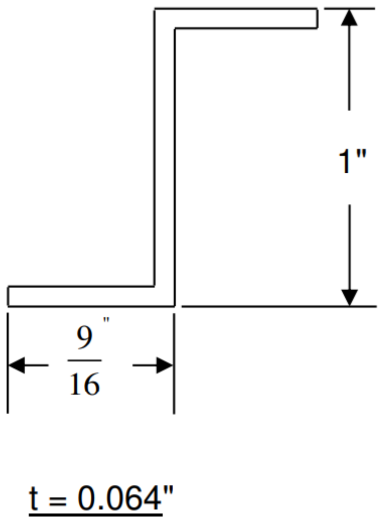
- **In fact, whenever the direction of the applied moment (regardless what that direction may be) aligns with one of the principal axes, we would be wise to draw our y and z axes along the principal axes**

- **We can then express σ_x in the simple "$\frac{My}{I}$" fashion!**

4.1: General Flexure Formula

Consider the cantilevered z-beam below with two bending couples as applied loads. Z-beams are often used as stiffeners in aircraft construction with the upper or lower surface riveted to the skin.





Find
(a) the normal stress variation over the cross-section and
(b) the equation of the neutral axis.
(c) Sketch the normal stress distribution.